https://www.sciencedirect.com/science/article/pii/S0749208117300190

Mammographic signs of screen-detected findings include

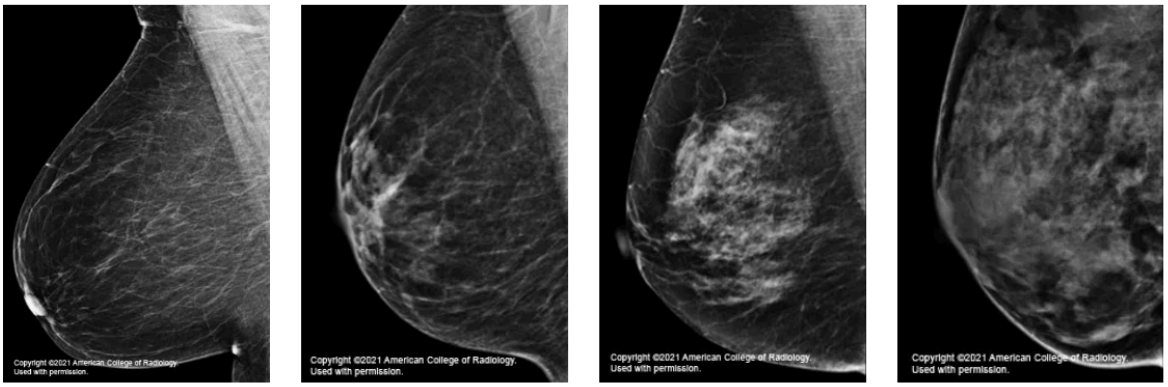
- lesions that are circular or oval, stellate or spiculated;
- micro-calcifications that may or may not relate to a tumor;
- abnormal skin thickening, asymmetries and combinations of these.

Presence of proportionately high volume of radiographic breast density is a risk factor [about density below]

https://www.cancer.org/cancer/types/breast-cancer/screening-tests-and-early-detection/mammograms/breast-density-and-your-mammogram-report.html

Breast density

Breasts that are extremely dense make it harder to see masses or other findings that may appear as white areas on the mammogram.



Category A: Breasts are almost all fatty tissue.

Category B: There are scattered areas of dense glandular and fibrous tissue (seen as white areas on the mammogram).

Category C: More of the breast is made of dense glandular and fibrous tissue (described as **heterogeneously dense**). This can make it hard to see small masses in or around the dense tissue, which also appear as white areas.

Category D: Breasts are **extremely dense**, which makes it harder to see masses or other findings that may appear as white areas on the mammogram.

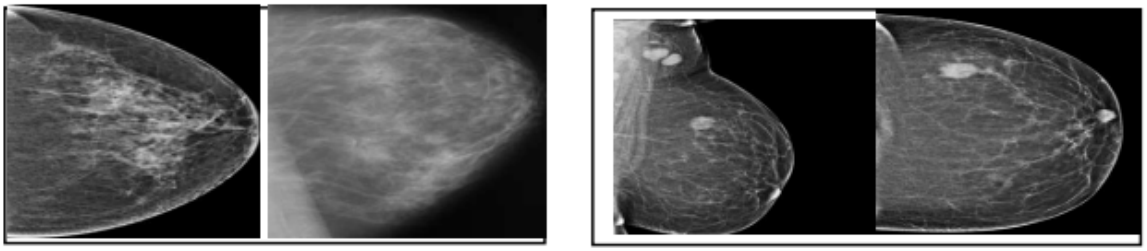
(patients whose breast density falls into categories C (heterogeneously dense) or D (extremely dense) are considered to have dense breast tissue. This includes about half of all women in the US who have mammograms to look for breast cancer.)

1. Women who have dense breast tissue have a higher risk of breast cancer compared to women with less dense breast tissue.
2. Dense breast tissue = white on a mammogram.

Breast masses and cancers = white
fatty tissue = almost black on a mammogram

https://www.sciencedirect.com/science/article/pii/S1877050925017107

- benign tumors typically exhibit smoother, well-defined edges
- malignant tumors often appear irregular and spiculated



A B

Fig 2: A) Benign Mammograms B) Malignant Mammograms.

GLCM technique takes gray mass and examines spatial relationships between adjacent pixels, considering their gray level intensities, distances, and orientations.